The Way Home in a Finite Scope

And How a Reader Turns a Search into a Route

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How to read the labels. A label records which slot a route actually landed on. **STANDS** marks an established theorem or measured fact, cited. **RUNS** marks a route executed in code; W1 to W5 name the block of paper1_witnesses.py whose full output is Appendix A. **DEFINITION** marks a term this paper fixes. **OPEN** marks a requirement written down whose route has not been run. **DEAD END** marks a route that ran and landed somewhere other than where it was aimed; it stays on the ledger and is routed back to its owner.

Abstract

A reader who needs anything must search. A reader who needs that can route. The two sentences share their first two words; the field they address is the same; the difference is the address the reader brings. This paper derives that difference from one held requirement and follows what the requirement permits, first from the side of the finished structure and then from the side of a reader inside a finite scope.

Held whole, the requirement that every state of an n-bit shift register has a way home admits exactly the laws in which the arriving bit is the oldest bit exclusive-or a function of the rest, 2^(2^(n−1)) of them, and the way home is the same law read n steps later. Weakened to what a reader actually has, a closed scope and a step that only goes forward, the requirement still forces a loop, and the loop is the one place the way home survives: on it the step is a bijection, and everything off it is history that has been used up. A run first repeats, tail and loop together, after about 2^(n/2 + 0.33) steps, half the register's bits, which is the birthday bound that sets collision cost at 2^128 for a 256-bit hash.

Between the two sides sits the reader's address. On the de Bruijn graph a reader with no address needs about 2^n random shifts to reach a target, and a reader with the address reaches every target in at most n. Each bit the reader brings halves what is left. A memory, in this paper's sense, is a stored constraint that turns a search into a route, and its size is the number of bits by which it shrinks the candidates. Lightning does this in plain sight: the first leader branches, a streamer from below selects one branch,

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and every later stroke runs down the lit channel without branching. The paper is written the same way, with the held end first and the code last, as witness that the path closes.

How the paper runs

Most derivations launch from a starting point and report where they land. This one holds the end fixed from the first line and asks, at each step, what that end still permits. It is a boundary-value problem rather than an initial-value one. One boundary is the reader's starting state, which is ignorance. The other is the held end, which the whole derivation answers to. The coding name for the method is Dijkstra's: derive the program from its postcondition, walking the requirement back one statement at a time [**STANDS**: Dijkstra 1976].

A lightning flash has the same shape. A stepped leader comes down from the cloud in jumps of tens of metres, branching through every opening in the air. An upward streamer rises from whatever stands closest on the ground and meets one branch. The instant the channel is complete, the return stroke flashes back up it, and it lights only the branch that connected; the others stay dark. Tens of milliseconds later a dart leader comes down the same channel without branching at all [**STANDS**: Rakov & Uman 2003]. Part I below is the leader from the top: what the held end permits. Part II is the streamer from the bottom: what a reader in a finite scope is forced into. Part III is the meeting. Part IV is the flash back, which lights the connected channel and leaves the dark branches out, except where one of them routed back to an owner and so belongs on the ledger.

The held end

Every state has a way home.

Hold that requirement and nothing else. Nothing is run yet. The question at every step is what the requirement permits, and the answer is already narrow before a single state is computed. Where the requirement is too strong for a reader to have, the paper weakens it once, to the scope the reader is actually in, and asks the same question again.

Part I. From the top: what the held end permits

1. The only laws that keep every way home

Take the n-bit feedback shift register. Its state is $(x_0, x_1, \ldots, x_{n-1})$; one step discards the oldest bit x_0, shifts the rest down, and appends $f(x_0, \ldots, x_{n-1})$. Write r for the $n - 1$ bits that survive the step. The held end says the step is a permutation: every state has exactly one predecessor.

Ask what that rules out. Suppose that for some r the law gives the same new bit whether the oldest bit was 0 or 1. Then the two states (0, r) and (1, r) step to the same successor, two states merge into one, and

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by counting some other state is left with no predecessor at all. The held end forbids it. So for every r the two values must differ, f(0, r) ≠ f(1, r), which is to say

$$f(x_0, r) = x_0 \oplus g(r), \quad \text{with } g \text{ any Boolean function of the other } n-1 \text{ bits.}$$

Conversely every law of that form keeps every way home, because from a successor (r, y) the oldest bit is recovered as $x_0 = y \oplus g(r)$. There are $2^{(2^n)}$ laws in all and $2^{(2^{(n-1)})}$ of this form. The requirement did the work: of the 65,536 four-bit laws it excluded 65,280 before any of them ran. [**STANDS**: Golomb 1967] The same count is K6 of the difference engine in the NEXUS Record (§ 123), recovered here from the held end. The exhaustive count at n = 2, 3, 4 gives 4 of 16, 16 of 256 and 256 of 65,536, with the permutations and the laws of this form coinciding exactly; 200 random laws of the form at each of n = 5 to 8 are permutations every time [**RUNS** W1].

2. The way home is the same law, read n steps later

The inverse needs no second machine. Written along the bit sequence the register produces, the forward law is $x_{t+n} = x_t \oplus g(x_{t+1}, \ldots, x_{t+n-1})$, and the way home is the same line solved for its other end:

$$x_t = x_{t+n} \oplus g(x_{t+1}, \ldots, x_{t+n-1}).$$

The bit written at step t is the bit consumed on the way home at step t + n. Writing and reading are one recurrence read at two positions n apart, never at the same instant. Run 200 steps out and then 200 steps home from the last n bits alone, and the register arrives at its seed, at every width tested from 5 to 8 [**RUNS** W2]. That fixes two words the rest of the paper uses. A WHAT is a write that carries its own inverse. A WHERE is the same bit read at the offset. There is no WHAT without a way home, and the way home is not stored beside the write; it is the write, read later.

Part II. From the bottom: what a reader in a finite scope is forced into

3. Weaken the end to what the reader has

A reader does not get a permutation. It gets a closed scope S, here 2^n states, and a step f that only goes forward, with no promise that two states never merge. Hold the weaker requirement: every run stays inside S. The question is the same: what does it permit?

It forces a repeat. A run that never leaves S must revisit a state within |S| steps, and from that state on it cycles forever. So every run ends on a loop [**STANDS**: elementary; Knuth 1997, §3.1]. Now subtract what the forward step accounts for and look at what remains. Let P be the states that lie on loops. The step maps P onto itself, and restricted to P it is a bijection: every state on a loop has exactly one predecessor on that loop. The way home has not disappeared. It has moved into the loop, the locally scoped region where the step can still be run backwards. Everything off the loop is a tail that feeds in and can never be returned to. Its history has been used up.

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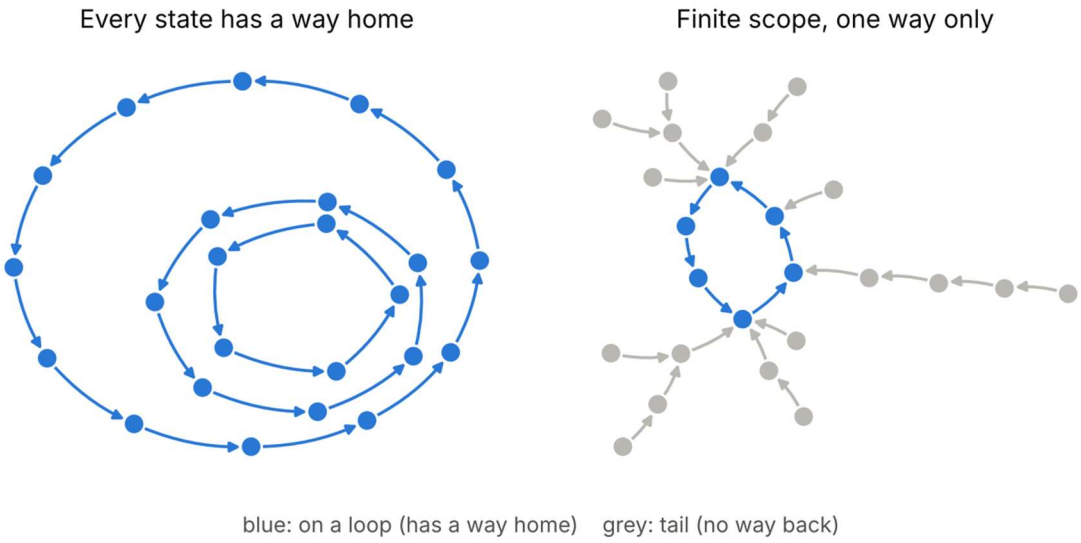


Figure 1. Left: a permutation on 24 states. Every state lies on a loop and has a way home. Right: a one-way map on 24 states. Only the loop (blue) keeps a way home; the tails (grey) feed in and cannot be returned to.

The count matches the theory of random mappings. Over 400 one-way maps on 256 states the loops held on average 19.5 states, against the expected √(πN/2) = 20.1; over 100 maps on 4,096 states, 74.3 against 80.2; over 30 maps on 65,536 states, 349.8 against 320.8. In every map the step restricted to its loops was a bijection, and in every permutation tested every state was on a loop [**RUNS** W3] [**STANDS**: Flajolet & Odlyzko 1990]. Single maps scatter widely because each has only a few components; the means converge.

So no way home does not mean a thing is running. It means it cannot stay where it is: it moves until a loop takes it, and inside the loop it has a way home again, a local one, because the loop comes back around to it. The running part is the loop. The tail is the instructions that were consumed to reach it. Read this way, decompilation and decomposition are the same pass: each returns a running loop to the field it came from, and each recovers the loop's structure but not the tail, because the tail is what was spent.

4. Where it can no longer go back: half the bits

The weakened requirement also fixes when a run first repeats, counting the tail and the loop together. For a one-way step on N states the expected number of steps before a run first repeats is √(πN/2) [**STANDS**: Flajolet & Odlyzko 1990]. For an n-bit register that is 2^(n/2 + 0.33): the point of no return sits at half the register's bits. Measured with 3,000 runs at each width from 8 to 32 bits, the mean grows from 19.7 steps at one byte to 83,216 steps at 32 bits, and its logarithm stays within 0.03 of n/2 + 0.33 at every width [**RUNS** W4].

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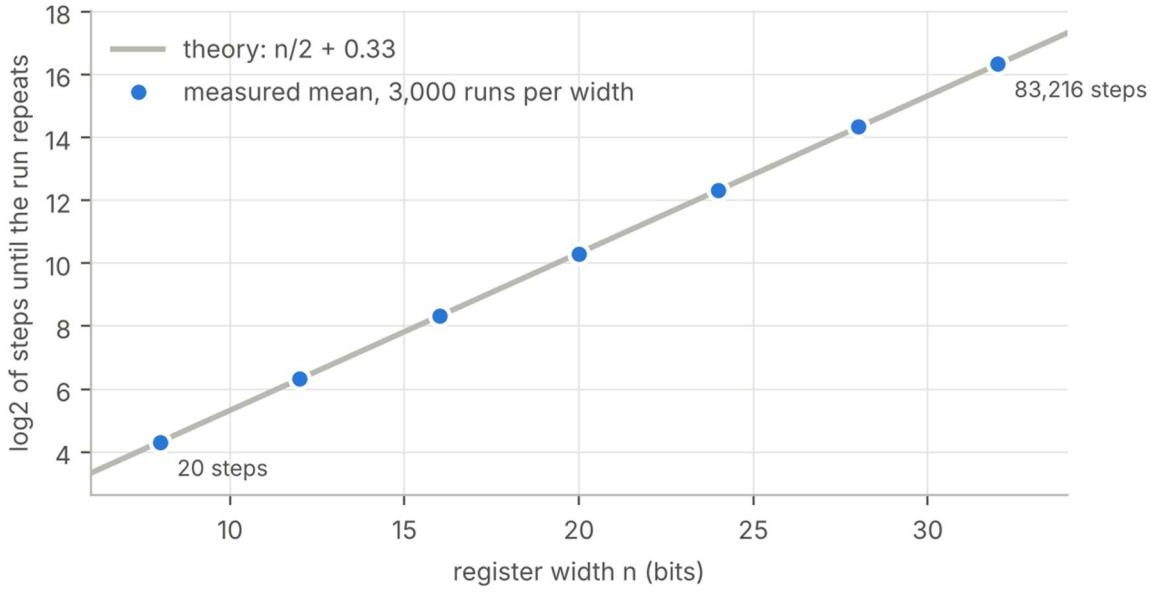


Figure 2. Mean number of steps before a one-way run repeats, against register width, with the theory line n/2 + 0.33. One byte first repeats after about 20 steps; 32 bits after about 83,000.

One byte stops being able to go back after about 20 steps. A 256-bit one-way step does so after about 2^128, which is the birthday bound that sets the cost of finding a collision in a 256-bit hash, and the loop shape is exactly what Pollard's rho method walks to find one [**STANDS**: Pollard 1975; van Oorschot & Wiener 1999]. The point where the tail enters the loop is where knowledge of origin ends and local running begins. In this program's own phrase, ignorance is the end of the compiler and the start of relative physics: past the merge, the reader holds a loop it can run in both directions and no route back to where the run began.

Base and width are different operations, and the distinction matters here. Changing the base in which a value is written re-reads the same value at a different address; it changes no capacity. Changing the register width changes the scope, and with it the number of steps before the loop closes. The byte layers of a scope matter as scopes, not as bases.

Part III. The meeting: I need that

5. The same field, two readers

Put the reader on the de Bruijn graph of order n, the graph whose 2^n nodes are the n-bit strings and whose edges are the shift: discard the oldest bit, append a new one. The field is complete; every node is reachable from every other. Two readers stand on it. The first needs anything: it has no address and appends random bits until it happens to stand on the target. The second needs that: it holds the target's n bits and shifts them in.

The first reader needs about 2^n steps. Measured over 400 walks at each width, the mean was 59.1 steps for 64 nodes, 253.9 for 256, 1,009.1 for 1,024, 4,111.8 for 4,096 and 16,444.4 for 16,384 [**RUNS** W5]. The

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second reader needs at most n. At n = 8 every one of the 65,536 ordered pairs of start and target routes in at most 8 shifts, the mean is 6.392 because a suffix of the start often already matches a prefix of the target, and 17,622 pairs need all 8 [RUNS W5] [STANDS: de Bruijn 1946; Samatham & Pradhan 1989].

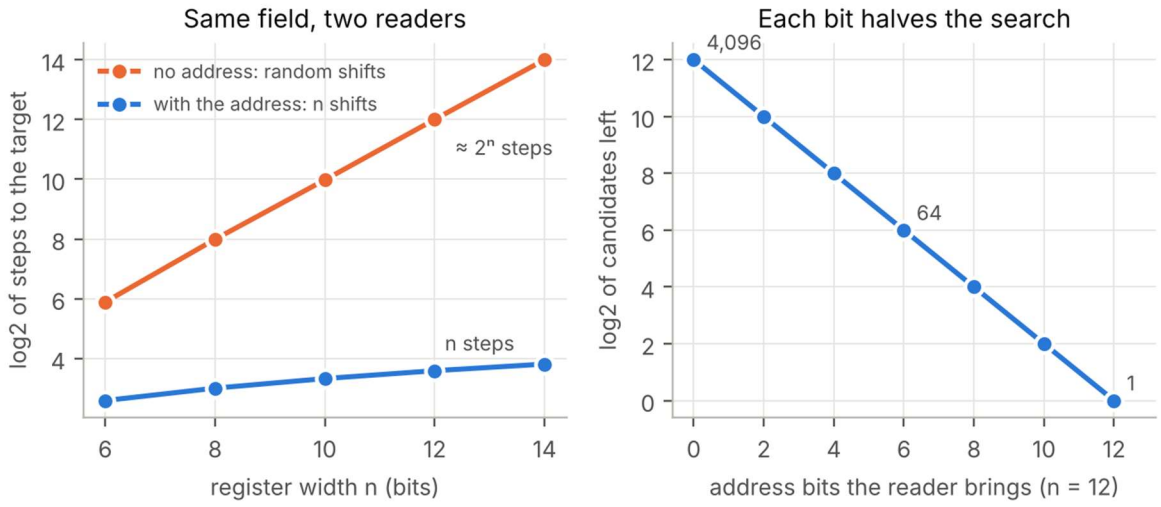


Figure 3. Left: steps to reach a target on the order-n de Bruijn graph, for a reader with no address (random shifts, about 2^n) and a reader with the address (n shifts). Right: at n = 12, each address bit the reader brings halves the candidates left, from 4,096 to 1.

Between the two readers lies every partial address. A reader holding k of the n bits has 2^(n–k) candidates left, and each further bit halves them; at n = 12 the candidates fall from 4,096 with no bits to 64 at six bits and to one at twelve, where the search has become a route of twelve steps [RUNS W5]. The words "I need" did not change between the two readers, and the field did not change. The search was removed by the reader meeting the field with an address, not by the field. The reader pays for specificity in bits, and n bits pay for all of it.

6. Memory is the channel

Return to the flash. The first leader's branches are the search. The streamer from the ground is the reader's constraint, rising from one local place and selecting one branch out of many. The return stroke lights only the connected channel. Then the dart leader comes down that channel without branching, because the channel is still there: the first stroke left a stored constraint in the air, and the next stroke routes along it instead of searching.

That gives this paper's working definition [DEFINITION]. A memory is a stored constraint that turns a search into a route. Its size is the number of bits by which it shrinks the candidates:

$$m = \log_2(N / C), \quad \text{where } N \text{ is the scope and } C \text{ the candidates the constraint leaves.}$$

A memory that leaves one candidate in a scope of 2^n states holds n bits, which is exactly the address, and the route it buys is n steps long. A memory that leaves sixty-four candidates in 4,096 holds six bits and leaves a search of sixty-four. Nothing in the definition says where the constraint is kept: in a channel of

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ionised air, in a register, in a table, or in the weights of a trained network. How to compute such a memory for an artificial system, and how to read one back out of it, is the next requirement this program writes down [OPEN].

7. The return stroke: how a landing is read back

A reader's claims about the field are holes: a WHAT written down before its WHERE is read. The protocol that keeps a hole honest has four steps in a fixed order. Declare the WHAT and the WHERE the route predicts, and seal the text with SHA-256 before any data is linked, including in the seal the readout through which the claim returns, its way home. Link the data, with any missing value marked unlinked; a slot with no data cannot land. Land, comparing the predicted WHERE with where the data puts the feature. Control, asking how often a random WHERE of the same size lands as well.

The hash is used here for what it is good for. It runs only forward, and no one can feasibly produce a different declaration with the same seal, so the seal fixes the WHAT before the data arrives and keeps either party from moving the WHERE afterwards. That is the reverse of asking a hash for a way home to its input, which the tail of a one-way scope does not provide.

A landing is read back the way a dependency-injection container resolves a request. The feature is bound when exactly one registered provider owns it. It is ambiguous when two providers claim it, and then an independent readout is the qualifier. It is unbound when no provider is registered, and then the container is naming the next WHAT to write. A DEAD END in this ledger is never an error in the code. The route ran and landed, and the landing belongs to something else; the work is to route it back to its owner. The protocol and its resolver are the companion code route_protocol.py and di_route_back.py, and the companion paper on atomic and nuclear shells runs them on measured data.

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Part IV. The flash back

8. The ledger

Table 1. What landed, and where.

Claim	Status	Witness
Every state keeps a way home exactly when $f = x_0 \oplus g(\text{rest})$; $2^{(2^{(n-1)})}$ laws	STANDS / RUNS	Golomb 1967; W1, exhaustive $n \leq 4$
The way home is the same law read n steps later	RUNS	W2, n = 5 to 8
A finite one-way scope forces a loop; only the loop keeps a way home	STANDS / RUNS	elementary; W3, 8 to 16 bits
A one-way n-bit scope repeats after about $2^{(n/2 + 0.33)}$ steps	STANDS / RUNS	Flajolet & Odlyzko 1990; W4, 8 to 32 bits
An address turns about 2^n random steps into at most n	STANDS / RUNS	de Bruijn routing; W5
Each address bit halves the candidates	RUNS	W5, n = 12
A memory is a stored constraint of $\log_2(N/C)$ bits	DEFINITION	Section 6
Computing and reading back such a memory in an AI system	OPEN	next requirement

9. Corrections carried into the protocol

Corrections, logged where they changed the rules

Missing is unlinked. An earlier electron-affinity check set missing values to zero and then tested whether the noble gases were at or below zero; the check could not fail. It was relabelled STANDS, as experiment, and the rule became: a slot with no data never counts as a landing.

Never cut a generator by hand. An earlier nuclear audit started its rule where the textbook says spin-orbit takes over, and missed the closure at 6 that the same rule writes when run from the start. A later declaration then stopped the generator before 168 and 184. The rule became: run the route uncut and state the landing domain separately.

The seal carries the way home. Early seals named the prediction but not the readout it returns through. From this paper on, the readout is part of the sealed text.

Name the object counted. A count of Madelung-group capacities, 2, 2, 8, 8, 18, 18, 32, 32, was once labelled as period lengths. The periods straddle the groups; their lengths, 2, 8, 8, 18, 18, 32, 32, inherit the pairing.

10. What this paper does not claim

The random-mapping numbers describe a typical one-way step. A particular structured step can close its loop sooner or later, and a hash function is a structured step; the birthday bound is its cost under the standard assumption that it behaves like a typical one. The loop gives a local way home and nothing more. A one-way scope offers no route back to the start of its tail, and this paper offers none; for a hash, the only legitimate uses of the structure here are the seal and collision cost. The definition of memory in

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Section 6 is a definition. Its value for artificial systems has to be shown by computing one, and that is OPEN. The lightning account is established atmospheric physics, used as an instance of the shape; nothing here claims that lightning computes.

11. The next stroke

The channel this paper lights is used by the next one. Working Paper 3, The Piece with No Way Home, follows it without branching: in every oscillator shell the one orbital with no pseudospin partner is the highest-j orbital, and it is the piece that leaves to close the shell below; the remainder of each shell is a copy of the shell before it; and the closure at 16 falls out of that chain instead of being searched for. Its landing on measured first-excited-state energies of even-even nuclei is already sealed and run.

12. Conclusion

One requirement was held from the first line. In its full form it admits only the laws whose arriving bit carries the leaving bit, and their way home is the same law read later. Weakened to a reader's closed scope, it still forces a loop, and the loop is where the way home goes: a bijection inside, a consumed tail outside, and a first repeat at half the bits. Between those two boundaries stands the reader, and the only thing that separates a search of 2^n from a route of n is the address the reader brings. A memory is that address, kept.

Acknowledgment. Computation and drafting were done with Claude (Anthropic) as a working partner. Every number marked RUNS is reproduced by the companion scripts paper1_witnesses.py and paper1_figures.py.

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Appendix A. Output of paper1_witnesses.py

The companion script recomputes every number marked RUNS in about four minutes, most of it the 32-bit runs of W4. Its complete output follows; the block labels match the tags in the text.

```
== W1  feedback laws with a way home for every state ==
 n=2:   16 laws, way home for every state   4, of form xo^g(rest)   4, both   4, 2^(2^(n-1)) = 4
 n=3:   256 laws, way home for every state  16, of form xo^g(rest)  16, both  16, 2^(2^(n-1)) = 16
 n=4:  65536 laws, way home for every state  256, of form xo^g(rest)  256, both  256, 2^(2^(n-1)) = 256
 n=5: 200 random laws of form xo^g(rest): permutations 200/200
 n=6: 200 random laws of form xo^g(rest): permutations 200/200
 n=7: 200 random laws of form xo^g(rest): permutations 200/200
 n=8: 200 random laws of form xo^g(rest): permutations 200/200


== W2  the way home is the same law read n steps later ==
 n=5: 200 steps out, 200 steps home from the last 5 bits alone: home = True
 n=6: 200 steps out, 200 steps home from the last 6 bits alone: home = True
 n=7: 200 steps out, 200 steps home from the last 7 bits alone: home = True
 n=8: 200 steps out, 200 steps home from the last 8 bits alone: home = True


== W3  one-way map on a finite scope: the way home survives only on the loop ==
 n= 8: 400 maps, mean loop states   19.5 of 256 (theory sqrt(pi N/2) =   20.1), restriction to the loop is a bijection in every map: True
 n=12: 100 maps, mean loop states   74.3 of 4096 (theory sqrt(pi N/2) =   80.2), restriction to the loop is a bijection in every map: True
 n=16:  30 maps, mean loop states  349.8 of 65536 (theory sqrt(pi N/2) =  320.8), restriction to the loop is a bijection in every map: True
 invertible maps (permutations), n = 8, 12, 16: every state is on a loop: True


== W4  where it stops being able to go back: half the bits ==
 n= 8: mean tail     9.4  loop     10.3  tail+loop     19.7  theory     20.1  log2 =  4.30  (n/2 + 0.33 =  4.33)
 n=12: mean tail    39.9  loop     40.4  tail+loop     80.3  theory     80.2  log2 =  6.33  (n/2 + 0.33 =  6.33)
 n=16: mean tail   160.3  loop    165.2  tail+loop    325.5  theory    320.8  log2 =  8.35  (n/2 + 0.33 =  8.33)
 n=20: mean tail   640.7  loop    630.2  tail+loop   1270.9  theory   1283.4  log2 = 10.31  (n/2 + 0.33 = 10.33)
 n=24: mean tail  2559.2  loop   2557.4  tail+loop   5116.5  theory   5133.6  log2 = 12.32  (n/2 + 0.33 = 12.33)
 n=28: mean tail 10221.6  loop  10496.4  tail+loop  20718.1  theory  20534.3  log2 = 14.34  (n/2 + 0.33 = 14.33)
 n=32: mean tail 41983.8  loop  41232.3  tail+loop  83216.1  theory  82137.2  log2 = 16.34  (n/2 + 0.33 = 16.33)


== W5  search vs route on the de Bruijn graph B(2,n) ==
 n=8: every one of 65536 (start, target) pairs routes in at most 8 steps; 17622 need all 8; mean 6.392
 n= 6: no address (random shifts): mean    59.1 steps to hit the target (N = 64);  with the address: 6
 n= 8: no address (random shifts): mean   253.9 steps to hit the target (N = 256);  with the address: 8
 n=10: no address (random shifts): mean  1009.1 steps to hit the target (N = 1024);  with the address: 10
 n=12: no address (random shifts): mean  4111.8 steps to hit the target (N = 4096);  with the address: 12
 n=14: no address (random shifts): mean 16444.4 steps to hit the target (N = 16384);  with the address: 14
 partial address, n = 12: k known bits leave 2^(12-k) candidates; expected candidates tried (2^(12-k)+1)/2
   k= 0: candidates  4096, expected tried  2048.5
   k= 2: candidates  1024, expected tried   512.5
   k= 4: candidates   256, expected tried   128.5
   k= 6: candidates    64, expected tried    32.5
   k= 8: candidates    16, expected tried     8.5
```

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k=10: candidates 4, expected tried 2.5

k=12: candidates 1, expected tried 1.0

each bit the reader brings halves the candidates; at k = n one remains and the route is n steps

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